

Unit 8

Categorical Data and Chi-square Tests

Lecture notes for IB DP Biology and Chemistry

Course purpose	To help students move from raw experimental data to cautious, justified scientific claims.
Unit focus	Counts, proportions, goodness-of-fit, chi-square independence tests, exact alternatives for small samples, and categorical effect sizes.
Central habit	Do not run chi-square on percentages alone. Use counts, expected counts, effect size, and cautious wording.

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1 Purpose of Unit 8

Many IB Biology and Chemistry investigations do not produce continuous measurements such as height, absorbance, or concentration. They produce **counts in categories**:

- offspring with dominant or recessive phenotypes;
- seeds that germinated or did not germinate;
- bacteria that survived or died after treatment;
- patients with or without a symptom;
- samples that fall into low, medium, or high contamination categories.

The statistical question is no longer simply about means. It is about whether the observed category counts are compatible with an expected pattern, or whether two categorical variables appear associated.

Core thesis

Categorical-data inference starts with counts. Percentages are useful for communication, but the evidence strength depends on how many observations produced those percentages.

By the end of this unit, students should be able to:

1. distinguish categorical counts from continuous measurements;
2. explain why chi-square tests use observed and expected counts;
3. perform a chi-square goodness-of-fit test;
4. perform a chi-square test of independence;
5. calculate degrees of freedom for common chi-square tests;
6. recognize when expected counts are too small for the chi-square approximation;
7. choose exact binomial or Fisher's exact tests for small samples;
8. report effect size for categorical data;
9. avoid overclaiming from significant and non-significant categorical tests.

2 Counts first: not percentages first

A categorical dataset records how many observations fall into each category. The same percentage can mean weak or strong evidence depending on the sample size.

Result	Count form	Percentage	Evidence strength
Small class demo	8 survived out of 10	80% survived	Very uncertain
Larger experiment	80 survived out of 100	80% survived	More stable
Large survey	800 survived out of 1000	80% survived	Much more precise

The percentage is the same, but the uncertainty is not the same.

Common mistake

Do not perform a chi-square test on percentages alone. A table with 80% and 50% is incomplete unless the underlying counts are known.

Biology example

A treatment group has 60% survival and a control group has 80% survival. This could mean:

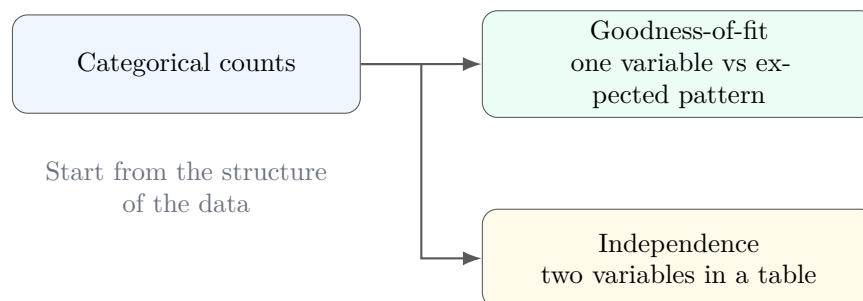
- 6/10 survived vs 8/10 survived;
- 60/100 survived vs 80/100 survived;
- 600/1000 survived vs 800/1000 survived.

The biological pattern is similar, but the statistical strength is very different.

3 Two main chi-square questions

There are two standard chi-square tests students usually meet first.

Test	Main question	Typical example
Chi-square goodness-of-fit	Does one categorical variable match an expected distribution?	Do offspring fit a 3:1 Mendelian ratio?
Chi-square independence test	Are two categorical variables associated?	Is survival associated with treatment group?



4 Observed and expected counts

The chi-square test compares observed counts with expected counts.

O_i = observed count in category i

E_i = expected count in category i under H_0

The test statistic is:

Chi-square statistic

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Each term measures how much one category contributes to the total mismatch between observation and expectation.

A large difference between observed and expected counts produces a larger chi-square statistic. The division by E_i matters: a difference of 10 is more serious when the expected count is 5 than when the expected count is 500.



5 Goodness-of-fit test: Mendelian 3:1 ratio

A **goodness-of-fit** test asks whether observed counts fit a proposed distribution, ratio, or model.

Research question

Do the offspring from a monohybrid cross fit the expected Mendelian 3:1 ratio?

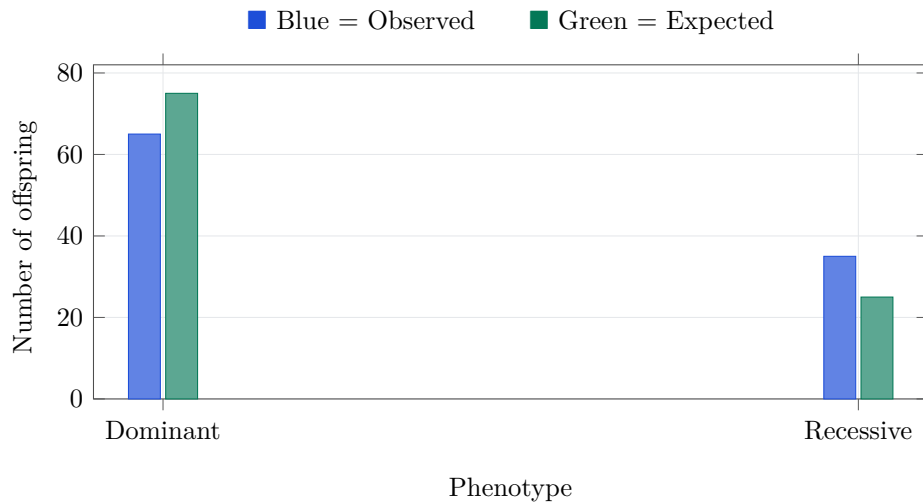
The hypotheses are:

H_0 : the offspring follow a 3:1 ratio.

H_A : the offspring do not follow a 3:1 ratio.

Suppose 100 offspring are observed:

Phenotype	Observed O	Expected proportion	Expected E	Contribution
Dominant	65	0.75	75	$(65 - 75)^2 / 75 = 1.33$
Recessive	35	0.25	25	$(35 - 25)^2 / 25 = 4.00$
Total	100	1.00	100	$\chi^2 = 5.33$



6 Degrees of freedom for goodness-of-fit

For a basic goodness-of-fit test with fixed expected proportions:

$$\text{df} = \text{number of categories} - 1$$

For the 3:1 example, there are two categories:

$$\text{df} = 2 - 1 = 1$$

The calculated statistic $\chi^2 = 5.33$ is then compared with a chi-square distribution with 1 degree of freedom. The p-value asks:

P-value wording

If the true offspring ratio were really 3:1, how often would random sampling produce a chi-square statistic at least as large as 5.33?

A small p-value means the observed counts would be unusual under the 3:1 model. It does not mean there is a small probability that Mendelian inheritance is true.

Interpretation warning

A goodness-of-fit test can show that data are difficult to reconcile with a proposed ratio. It does not identify the biological cause. Possible causes include sampling variation, small sample size, scoring error, linkage, viability differences, contamination, or an incorrect genetic model.

7 Expected counts and the small-sample problem

The chi-square test uses an approximation. That approximation becomes unreliable when expected counts are too small.

A common classroom rule is:

Expected-count rule of thumb

Expected counts should usually be at least 5 in each category. If expected counts are smaller, consider an exact test or simulation-based method.

This rule is not a law of nature, but it is a useful safety check for IB-level analysis.

Expected ratio	Smallest expected proportion	Bare minimum for expected count ≥ 5
3:1	$1/4 = 0.25$	$n \geq 20$
9:3:3:1	$1/16 = 0.0625$	$n \geq 80$
1:1:1:1	$1/4 = 0.25$	$n \geq 20$
1:2:1	$1/4 = 0.25$	$n \geq 20$

Quick check

A dihybrid cross is expected to produce a 9:3:3:1 ratio. A student obtains 48 offspring. Is a chi-square test automatically appropriate? Explain using expected counts.

8 Test of independence: treatment and survival

A chi-square test of independence asks whether two categorical variables are associated.

Research question

Is bacterial survival associated with treatment group?

The null and alternative hypotheses are:

H_0 : survival and treatment group are independent.

H_A : survival and treatment group are associated.

Suppose the observed counts are:

Group	Survived	Died	Total
Control	80	20	100
Antibiotic	50	50	100
Total	130	70	200

The expected count in each cell is calculated as:

Expected cell count

$$E = \frac{\text{row total} \times \text{column total}}{\text{grand total}}$$

For control survivors:

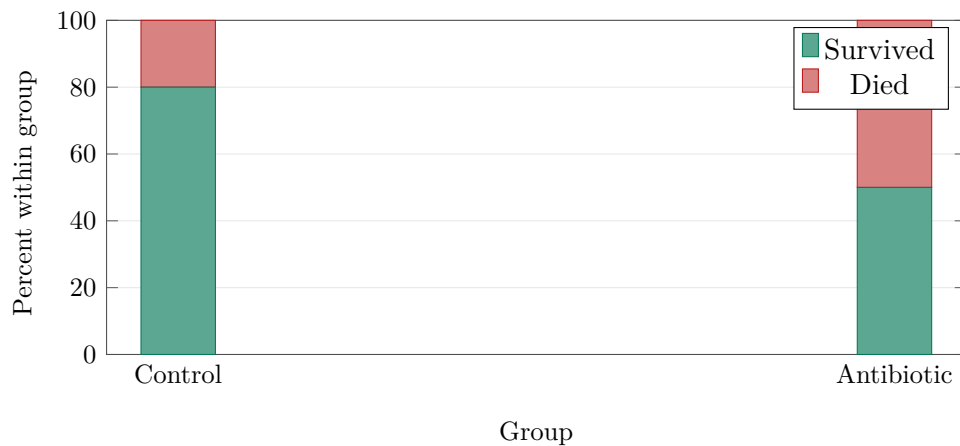
$$E = \frac{100 \times 130}{200} = 65$$

The complete expected table is:

Group	Expected survived	Expected died
Control	65	35
Antibiotic	65	35

9 Visualizing association in a 2 by 2 table

Percentages help describe the effect, but the test still uses counts.



The survival rate changed from 80% in the control group to 50% in the antibiotic group. The chi-square test asks whether this departure from independence is larger than expected by random sampling under the null model.

For an $r \times c$ independence table:

$$df = (r - 1)(c - 1)$$

For a 2 by 2 table:

$$df = (2 - 1)(2 - 1) = 1$$

10 Effect size for categorical data

The p-value tells whether the data are unusual under the null model. It does not tell the biological size of the effect.

For categorical data, report an effect size as well.

Effect size	Useful for	Meaning
Difference in proportions	Two groups	Absolute change, e.g. survival decreased by 30 percentage points.
Relative risk	Cohort or treatment studies	Risk in one group divided by risk in another group.
Odds ratio	Case-control studies and logistic models	Odds in one group divided by odds in another group.
Phi coefficient ϕ	2 by 2 tables	Correlation-like association measure for binary variables.
Cramer's V	Larger contingency tables	General association measure from 0 to 1.
Cohen's w	Goodness-of-fit	Size of deviation from expected proportions.

For the treatment-survival example:

$$\text{death risk in control} = \frac{20}{100} = 0.20$$

$$\text{death risk in antibiotic} = \frac{50}{100} = 0.50$$

$$RR = \frac{0.50}{0.20} = 2.5$$

So a clearer biological statement is:

Effect-size statement

The death risk was 2.5 times higher in the antibiotic group than in the control group in this sample.

This statement is more informative than simply saying $p < 0.05$.

11 Phi, Cramer's V, and Cohen's w

For a 2 by 2 table, the phi coefficient is:

$$\phi = \sqrt{\frac{\chi^2}{n}}$$

For larger contingency tables, use Cramer's V :

$$V = \sqrt{\frac{\chi^2}{n(k-1)}}$$

where k is the smaller number of rows or columns.

For goodness-of-fit, Cohen's w can be written as:

$$w = \sqrt{\sum \frac{(p_{\text{observed},i} - p_{\text{expected},i})^2}{p_{\text{expected},i}}}$$

Cutoff warning

Do not treat “small”, “medium”, and “large” effect-size labels mechanically. A small statistical association may still matter biologically if the outcome is serious or the exposure is widespread.

12 Exact tests for small samples

When expected counts are small, exact tests are often better than the chi-square approximation.

Situation	Better method
Two-category goodness-of-fit, small sample	Exact binomial test
More than two expected categories, small sample	Exact multinomial test or simulation
2 by 2 contingency table with small expected counts	Fisher’s exact test
Larger sparse table	Exact or Monte Carlo method

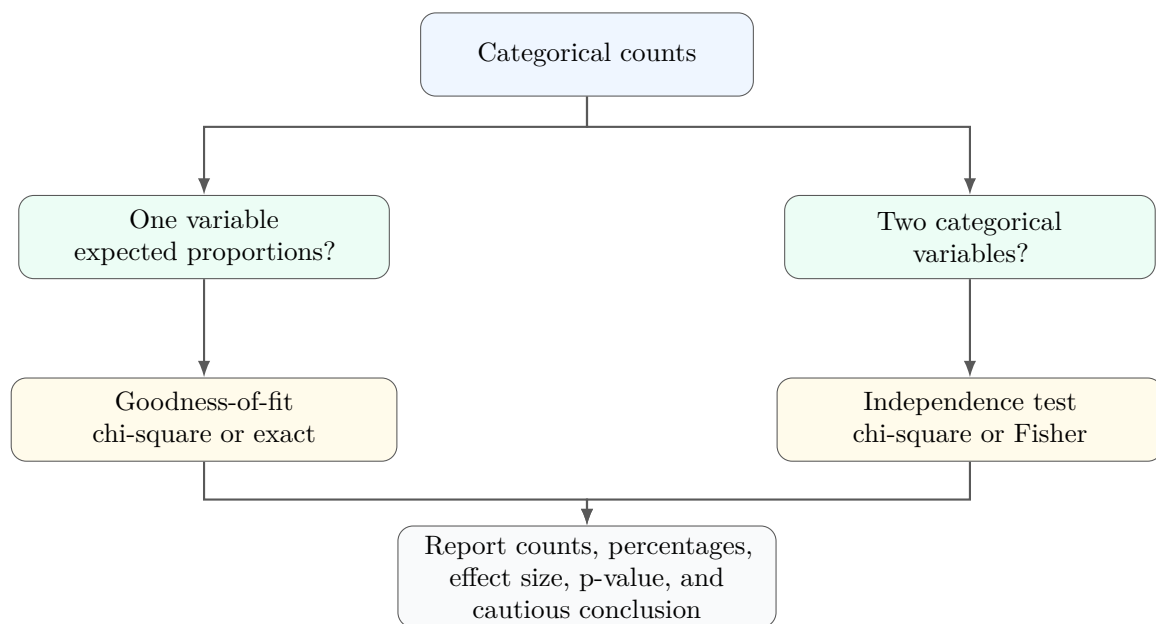
Small Mendelian sample

A 3:1 ratio predicts recessive probability $p = 0.25$. If only 12 offspring are observed, the expected recessive count is $12 \times 0.25 = 3$. This is below 5, so an exact binomial test is more appropriate than a chi-square approximation.

Exact methods do not magically solve poor design. They only avoid using a poor large-sample approximation.

13 Choosing the method

Use the data structure first. Do not choose the test by habit.



14 Writing categorical-data conclusions

A strong conclusion combines the observed pattern, the statistical evidence, and the effect size.

Weak wording

Avoid:

- “There was no association.”
- “The treatment caused survival to change.”
- “The null hypothesis is true.”
- “The result is not due to chance.”

Better wording

Use:

- “No statistically significant association was detected.”
- “The data provide evidence of an association between treatment and survival.”
- “The result would be unusual if treatment and survival were independent.”
- “The effect size suggests a biologically meaningful difference, but the design limits causal interpretation.”

Model conclusion

In the sample, survival was lower in the antibiotic group than in the control group, and death risk was 2.5 times higher in the antibiotic group. A chi-square test of independence indicated that the observed table would be unlikely if treatment group and survival were independent. This supports an association between treatment and survival, but causal interpretation depends on the experimental design.

15 Practice tasks

Task 1: Goodness-of-fit

A monohybrid cross is expected to produce a 3:1 ratio. A student observes 72 dominant and 28 recessive offspring.

1. Calculate the expected counts.
2. Calculate χ^2 .
3. State the degrees of freedom.
4. Write a cautious conclusion template without claiming proof.

Task 2: Independence

A germination experiment gives the following counts:

Condition	Germinated	Not germinated
Control	34	6
Salt solution	22	18

1. What are the two categorical variables?
2. Calculate the row and column totals.
3. Calculate the expected count for Control-Germinated.
4. Which effect size would communicate the result clearly?

16 Answer guide

Task 1

Total offspring:

$$72 + 28 = 100$$

Expected counts under 3:1:

$$E_{\text{dominant}} = 75, \quad E_{\text{recessive}} = 25$$

Chi-square statistic:

$$\chi^2 = \frac{(72 - 75)^2}{75} + \frac{(28 - 25)^2}{25} = \frac{9}{75} + \frac{9}{25} = 0.12 + 0.36 = 0.48$$

Degrees of freedom:

$$\text{df} = 2 - 1 = 1$$

Cautious conclusion:

The observed counts are close to the expected 3:1 ratio. The chi-square statistic is small, so the data do not provide strong evidence against the 3:1 model. This does not prove that the true ratio is exactly 3:1; it only means that this sample is compatible with that expectation.

Task 2

The two categorical variables are condition and germination outcome.

Totals:

Condition	Germinated	Not germinated	Total
Control	34	6	40
Salt solution	22	18	40
Total	56	24	80

Expected Control-Germinated:

$$E = \frac{40 \times 56}{80} = 28$$

A useful effect size would be the difference in germination proportions:

$$\frac{34}{40} - \frac{22}{40} = 0.85 - 0.55 = 0.30$$

So the control group had a germination rate 30 percentage points higher than the salt-solution group in this sample.

17 Unit 8 summary

Take-home message

Categorical-data analysis is not about means. It is about counts, expected counts, proportions, and associations. A chi-square p-value is useful only when the data structure and expected-count assumptions are appropriate. A good conclusion reports the observed pattern, the statistical evidence, the effect size, and the uncertainty in interpretation.

Question	Best first move
Does this fit an expected ratio?	Use goodness-of-fit thinking.
Are two categorical variables associated?	Use an independence-table approach.
Are expected counts small?	Consider exact or simulation methods.
Is the p-value significant?	Still report effect size and avoid overclaiming.
Is the p-value non-significant?	Say no significant association was detected, not that no association exists.